

Daniel Lane

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Summary

PhD in Mechanical Engineering specializing in real-time sensor fusion, autonomous perception, and embedded systems for uncrewed platforms. Dissertation research centered on LiDAR-camera fusion for maritime autonomy, with additional work in data-driven nonlinear dynamics. Career arc spans enterprise IT systems leadership, graduate research, international robotics competition, and university teaching. Seeking R&D roles in robotics, perception, aerospace autonomy, or applied AI.

Education

Embry-Riddle Aeronautical University – PhD in Mechanical Engineering Dec 2025
Dissertation: "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels"

Embry-Riddle Aeronautical University – MS in Mechanical Engineering May 2021

Stetson University – BS in Physics May 2009

Core Engineering Competencies

Software Development & Project Management: Python, C, C++, HTML/CSS/JavaScript, ROS/ROS2, Git

Machine Learning & Computer Vision: TensorFlow, PyTorch, OpenCV, YOLO, neural network architectures, transfer learning, reinforcement learning

Software & Tools: MATLAB/Simulink, NumPy/Pandas, Gazebo/RViz, Fusion 360, SolidWorks, 3D printing

Robotics & Perception: Sensor fusion, object detection/classification, camera/LiDAR calibration and time synchronization, GNSS/INS, embedded real-time systems, microcontrollers, navigation and path planning

Modeling & Control: Nonlinear dynamic systems, system identification, Kalman filters, information-theoretic methods, PID/LQR control

Engineering Projects & Research

LiDAR-Camera Fusion for Maritime Autonomy

ASV Perception Research, Robotics & Autonomous Systems Lab (RASL), ERAU | NEEC/ONR Grant 2024 – 2025

- Developed lightweight late-fusion method for multi-modal object detection on embedded platforms (NVIDIA Jetson AGX Xavier), generalizable across sensor modalities and detection frameworks
- Designed and implemented custom GStreamer/RTP pipeline embedding frame-accurate timestamps as SEI metadata, achieving sub-millisecond synchronization between Livox Horizon LiDAR and Sony IMX490 HDR camera streams
- Performed full spatial calibration (intrinsic and extrinsic) across a 7-sensor array (3 LiDAR, 3 4K cameras, 1 HDR camera), enabling unified point cloud and image fusion at >15 fps on embedded hardware
- Collected and processed approximately 4 TB of multi-modal sensor data over a 6-day on-water test campaign

Maritime RobotX Challenge & RoboBoat Challenge – Lead Perception Engineer

Sr. Software Team Member, Team Minion, ERAU 2024 – 2025

- Led perception system architecture for international ASV competition; team placed 3rd overall and 1st in technical documentation at the 2024 Maritime RobotX Challenge
- Developed real-time LiDAR-HDR fusion pipelines for object detection, classification, and color identification supporting autonomous mission planning
- Designed ground control interfaces and sensor pipelines for real-time embedded deployment under competition constraints

Consultation for Yamaha - Future Boat Concept

Co-Researcher 2024

- Collaborated under NDA with Yamaha Marine Division to develop high-level design concepts for future vessels based on feasibility analysis of emerging marine technologies

Data-Driven Modeling of Traffic Systems

Researcher, Complex Dynamical Systems Lab (CDSL), ERAU | NSF Grant

2022 – 2023

- Developed framework for recovering nonlinear system dynamics by detecting model structure through data-driven and information-theoretic methods
- Supervised undergraduate team on hardware integration and software testing for experimental validation

Professional Experience

Visiting Assistant Professor, ERAU – Daytona Beach, FL

Spring 2026

- Designed and deployed a custom agentic assistant to manage student communication: monitored emails, maintained anonymized conversation context, and generated LLM-assisted reply drafts with SMS alerts – sustaining sub-2-hour response time across 140 students for the full semester
- Integrated Canvas API to provide real-time visibility into assignment and submission status, enabling intelligent prioritization of grading and course preparation across 4 concurrent sections
- Taught 4 sections (140 students total) across Dynamics, Uncrewed Systems, and Mechatronics; developed original materials for Dynamics and modernized inherited curricula for the remaining courses

Imaging Systems Specialist, ERAU – Daytona Beach, FL

2017 – 2020

- Led OCR-based document workflow automation, coordinating between internal stakeholders and an external development team
- Integrated admissions and registrar systems to streamline student document processing, reducing manual work from hours to minutes

Technology Project Manager, FDOT - Turnpike – Ocoee, FL

2014 – 2017

- Managed development of a GIS-integrated road inspection dashboard unifying multiple legacy systems for statewide per-mile visual inspection, photo history, and maintenance tracking
- Coordinated bi-weekly development sprints with a third-party firm and a partner PM, managing requirements and delivery milestones

Senior Technical Roles

2009 – 2014

Apple Inc., Stetson University

- Managed enterprise IT infrastructure, technical troubleshooting, and multimedia systems across multiple organizations
- Built foundational expertise in systems integration, documentation, and automation workflows

Publications & Presentations

- D. Lane, "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels" (2025). Doctoral Dissertations and Master's Theses. 937. <https://commons.erau.edu/edt/937>.
- D. Lane and S. Roy, "Validating a data-driven framework for vehicular traffic modeling," *Journal of Physics: Complexity*, vol. 5, no. 2, p. 025008, May 2024. doi: 10.1088/2632-072X/ad3ed6.
- D. Lane and S. Roy, "Using information theory to detect model structure with application in vehicular traffic systems," *IFAC-PapersOnLine*, vol. 56, no. 3, pp. 367–372, 2023. 3rd Modeling, Estimation and Control Conference (MECC 2023). doi: 10.1016/j.ifacol.2023.12.051.
- D. Lane and S. Roy, "Creating Data-Driven Model Discovery Framework with Applications in Traffic Modeling," poster presentation, SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR, May 2023.
- D. Lane and E. Coyle, "Appendix D: Vision System," in *Minion: Design and Competition Strategy for the 2024 Maritime RobotX Challenge*, Team Minion Technical Design Report, RoboNation, 2024. Available: robonation.org [p. 30]